

A System of Multiple AI Agents for Automating Tokamak-Plasma Simulation for Nuclear Fusion Energy.

A hierarchical multi-agent PETSc system: from a plain-language prompt to verified, real-machine Grad-Shafranov equilibria.

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1. Motivation

Building correct, fast HPC simulation code demands scarce, implicit expertise. Can a system of multiple AI agents automate the path from a plain-language idea to verified code for a real fusion problem?

We answer yes for the tokamak Grad-Shafranov equilibrium, with verification as a first-class deliverable.

HPC library used is PETSc (Portable, Extensible Toolkit for Scientific Computation).

2. The problem being solved

Axisymmetric ideal-magnetohydrodynamics force balance for the poloidal magnetic flux $\psi(r, z)$:

$$\frac{\partial^2 \psi}{\partial r^2} - \frac{1}{r} \frac{\partial \psi}{\partial r} + \frac{\partial^2 \psi}{\partial z^2} = -\mu_0 r^2 \frac{dp}{d\psi} - \frac{1}{2} \frac{dF^2}{d\psi}$$

where p is plasma pressure, μ_0 is magnetic permeability, $F = rB_\theta$, B is magnetic field, and (r, θ, z) are cylindrical coordinates.

Sets the flux surfaces, magnetic axis, plasma shape, and safety factor q .

Solov'ev profiles admit an exact solution → a rigorous verification anchor.

3. The system of multiple AI agents

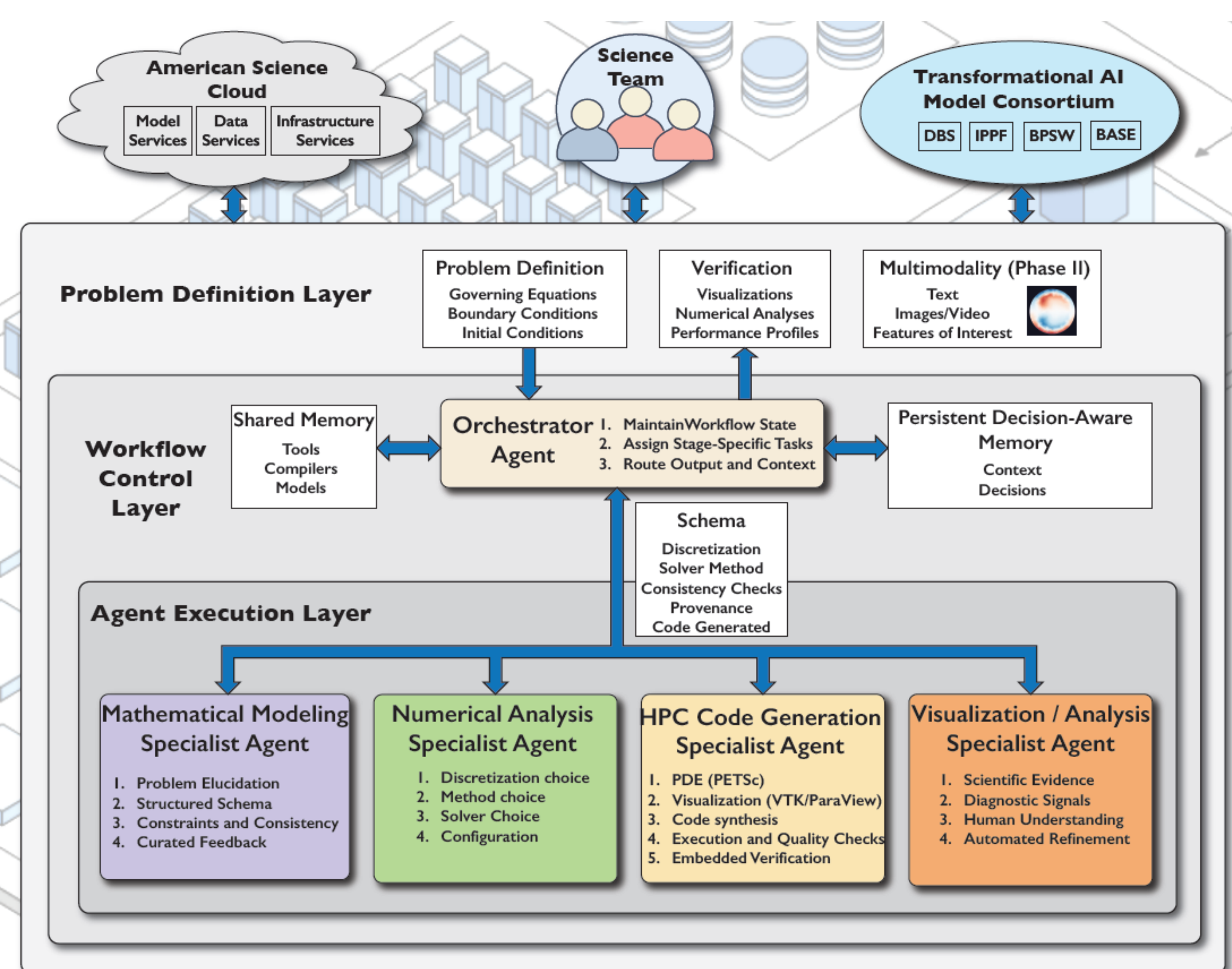


Figure 1: A Problem-Definition layer, an Orchestrator agent + shared / persistent memory (Workflow Control), and specialist Modeling / Numerical-Analysis / Code-Generation / Visualization agents (Agent Execution).

4. The pipeline in action

Prompt: “the Grad-Shafranov equilibrium for a tokamak plasma.”

Modeling → ‘Grad-Shafranov equation’, steady.

Numerical Analysis → non-linear ⇒ SNES (Scalable Nonlinear Equations Solvers) on a DMDA (Distributed Management Distributed Array).

Code Generation → one 252-line parametrized PETSc solver.

Compiled and ran on 1 and 4 MPI ranks without any human edits.

5. Generated solver (excerpt)

```
/* GS operator (normalized x=R/R0), 2nd-order FD */
DMDACreate2d(...,DMDA_STENCIL_STAR,...,&da);
dxx=(u[j][i+1]-2*u[j][i]+u[j][i-1])/hx2;
dyy=(u[j+1][i]-2*u[j][i]+u[j-1][i])/hy2;
f[j][i]=dxx-dx/x+dyy-((1-A)*x+A);
/* Dirichlet BC = analytic CF psi (nonzero) */
f[b]=u[b]-PsiExact(&u,x,y);
SNESetJacobian(snes,J,J,FormJacobian,&u);
SNESolve(snes,NULL,X);
```

6. Real-machine equilibria

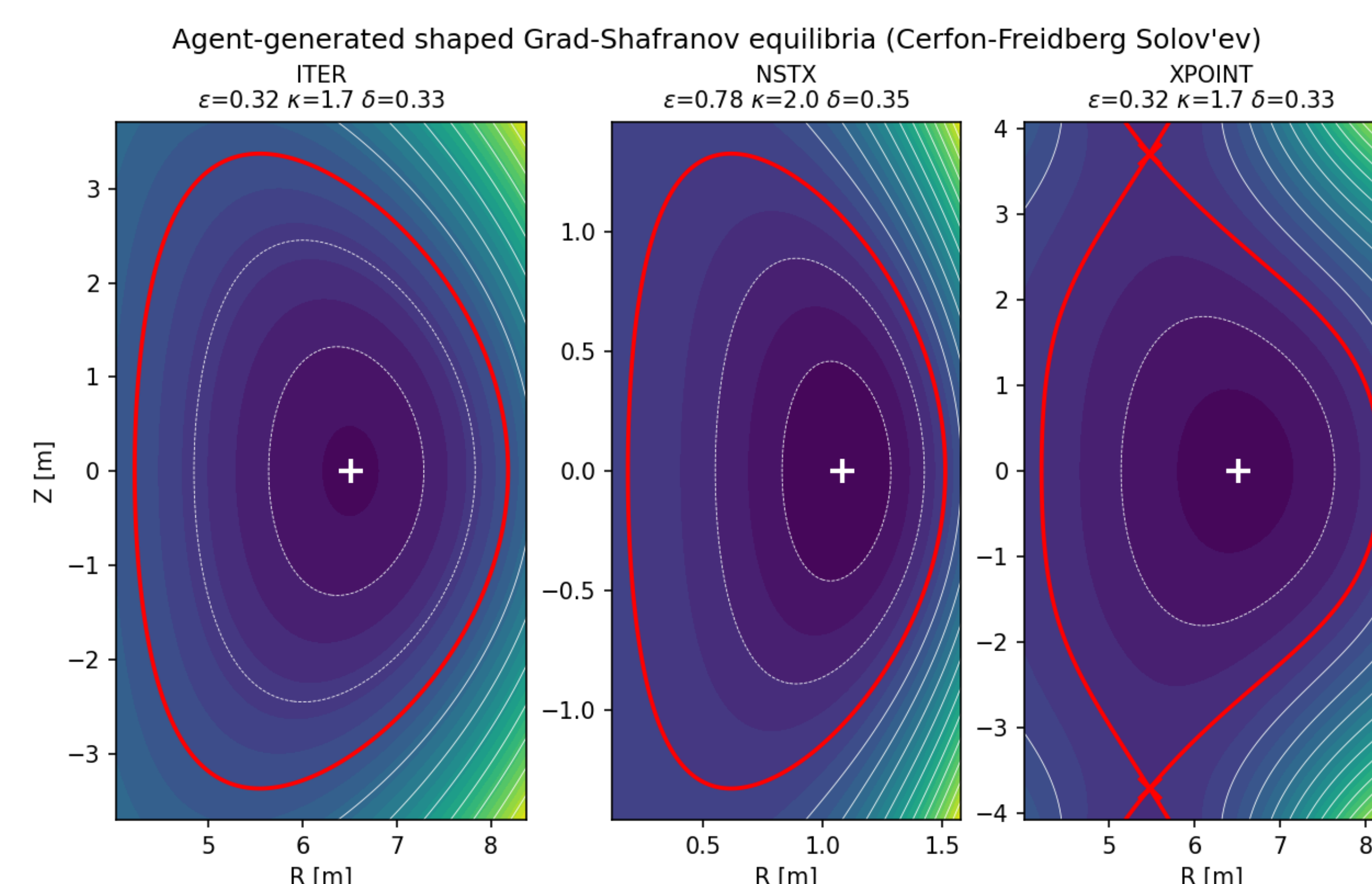


Figure 2: One agent-generated solver → ITER, spherical-tokamak (NSTX-like) and diverted double-null (X-point) equilibria. Red = separatrix; x = X-points.

7. Verification: second order

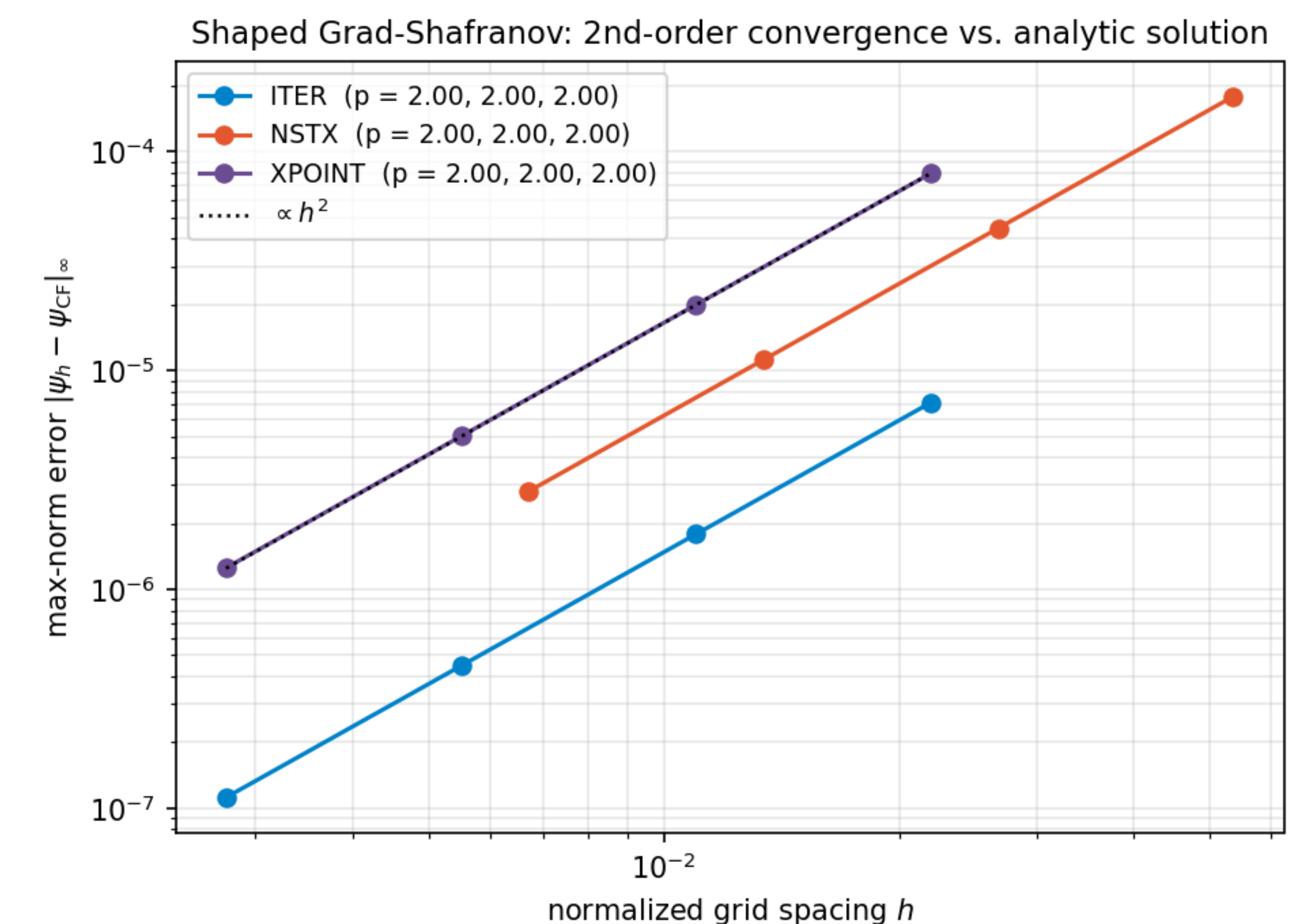


Figure 3: Method of manufactured solutions vs the exact Cerfon-Freidberg solution: observed order $p = 2.00$ for all three machines.

8. Safety factor and FreeGS cross-check

Safety-factor $q(\psi_N)$ computed from the verified flux surfaces:

ITER $q_{95} \approx 2.9$ · NSTX $q_{95} \approx 12.4$ · X-point $q_{95} \approx 3.2$

Cross-checked vs FreeGS’s independent q on the SAME field → agree to less than 0.2 %.

Measured κ, δ match the input shaping to around 1 to 2 %.

9. Results and conclusions

One 252-line solver, 0 hand-written, verified on 3 real-machine equilibria.

Second-order accurate ($p = 2.00$) on every case; runs on 1 and 4 MPI ranks.

Efficiency: wall-clock time of 311.4 s; around 27 LLM completions.

Verification-driven: exact-solution convergence + independent q cross-check.

github.com/engineer-scientist/petsc_mcp_servers_tokamak